
STATISTICAL ANALYSIS AND CLASSIFICATION OF THE ADULT INCOME DATASET

Adult Income dataset (1994 US Census)

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github.com/marizxv/t4e_DataScience_school

ADULT INCOME: 1994 US CENSUS DATA

Adult Income, 1994 US census: 48,842 people
(adult.data 32,561 + adult.test 16,281).

15 variables (6 numerical, 9 categorical).

Target: income >50K.

We kept the original train/test split.

24.9% of people in train earn more than \$50K.

Type	Variables
Numerical	Age, education num, hours per week, capital gain, capital loss, fnlwgt.
Categorical	Workclass, education, marital status, occupation, relationship, race, sex, native country.
Target	Income: more or less than \$50K per year.

DATA PREPROCESSING

Everything that learns from the data is fitted on train and only applied to test (to avoid data leak)

Cleaning

- Removed the dot in test labels (>50K. → >50K).
- Deleted rows with missing values: 2 399 train (7.37%), 1 221 test (7.50%)
- Deleted duplicates: 23 train, 5 test
- Left: 30 139 train, 15 055 test

New features

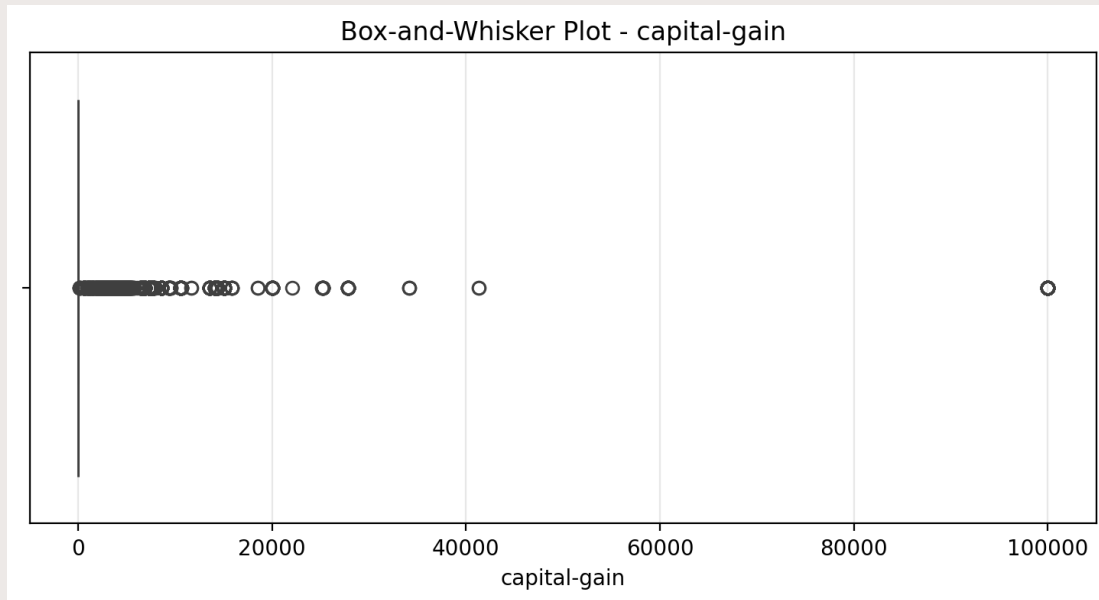
- has-capital, is-married, is_us, capital-net
- age-group (e.g. 17–24)
- hours-category (part-time, full-time, overtime)

Encoding

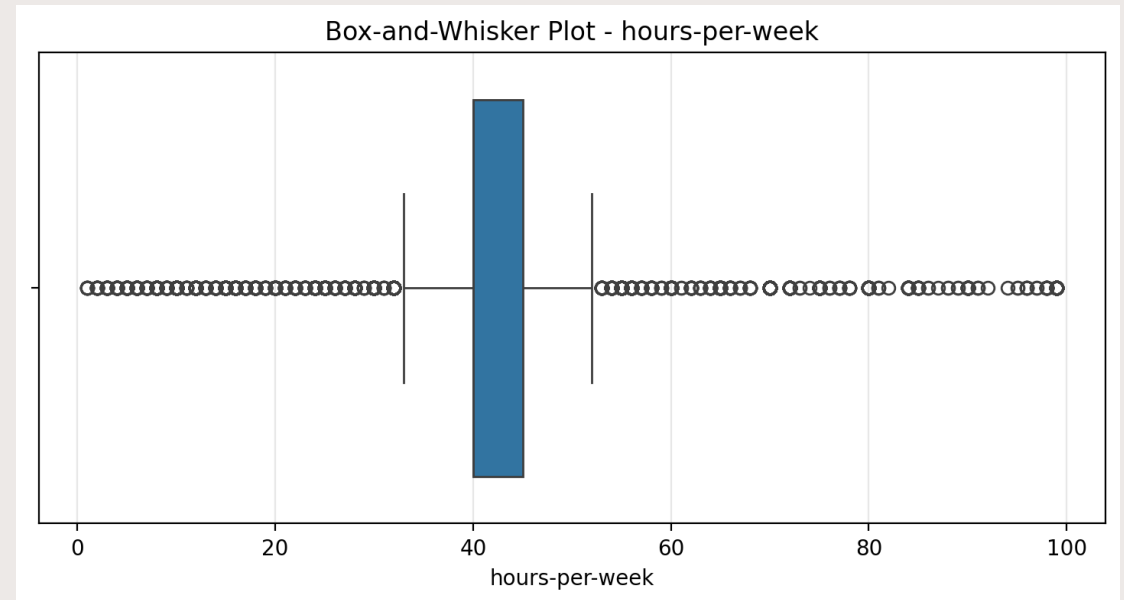
- sex 0/1, native-country → is_us
- One-hot: workclass, marital-status, occupation, relationship, race
- Dropped education (same as education-num) and fnlwgt
- Numerical standardized → 42 columns

OUTLIERS

- Found with z-score and IQR and kept, because they were representing real people, not errors
- capital-gain and capital-loss: $\log(1 + x)$, skewness of capital-gain $11.9 \rightarrow 3.07$



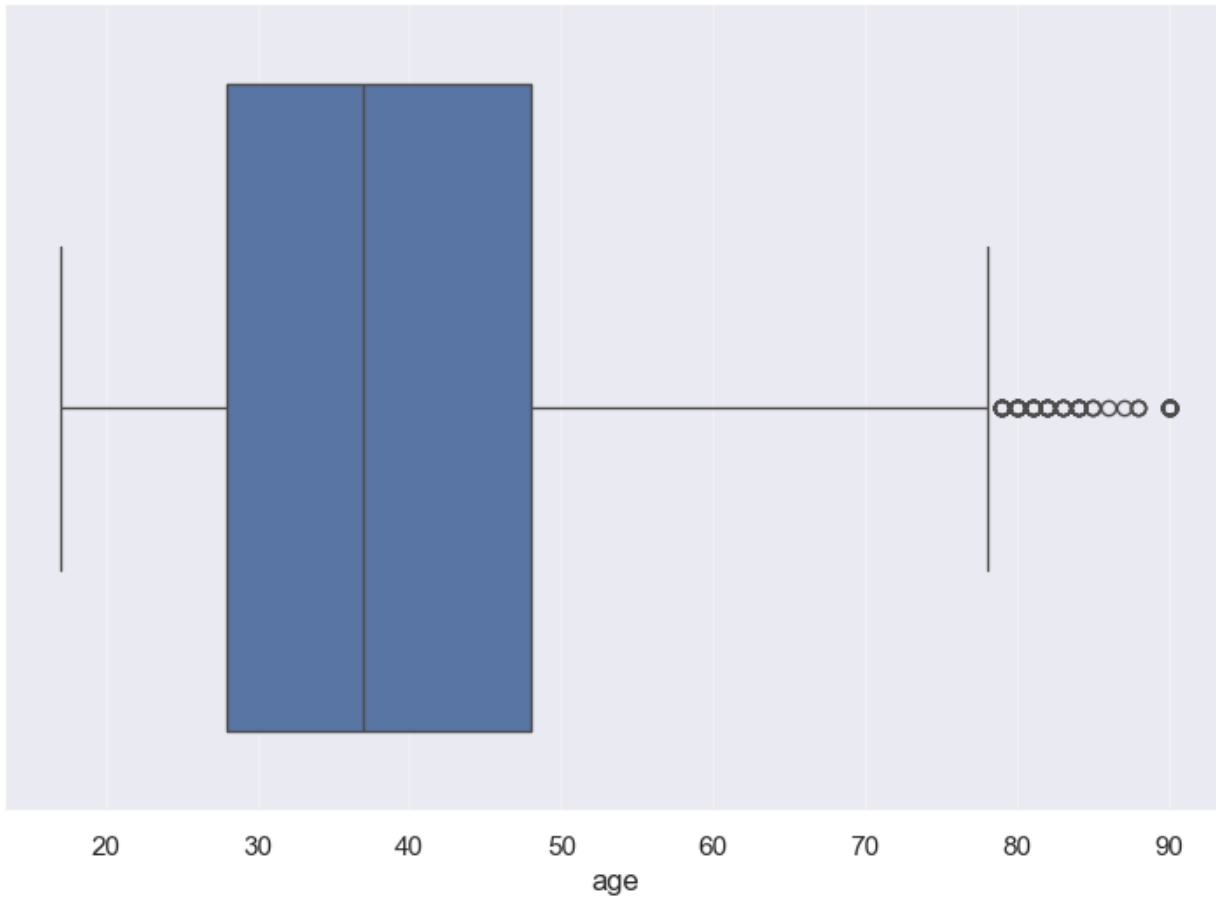
91.6% have exactly 0: no box, every non-zero value is an outlier dot



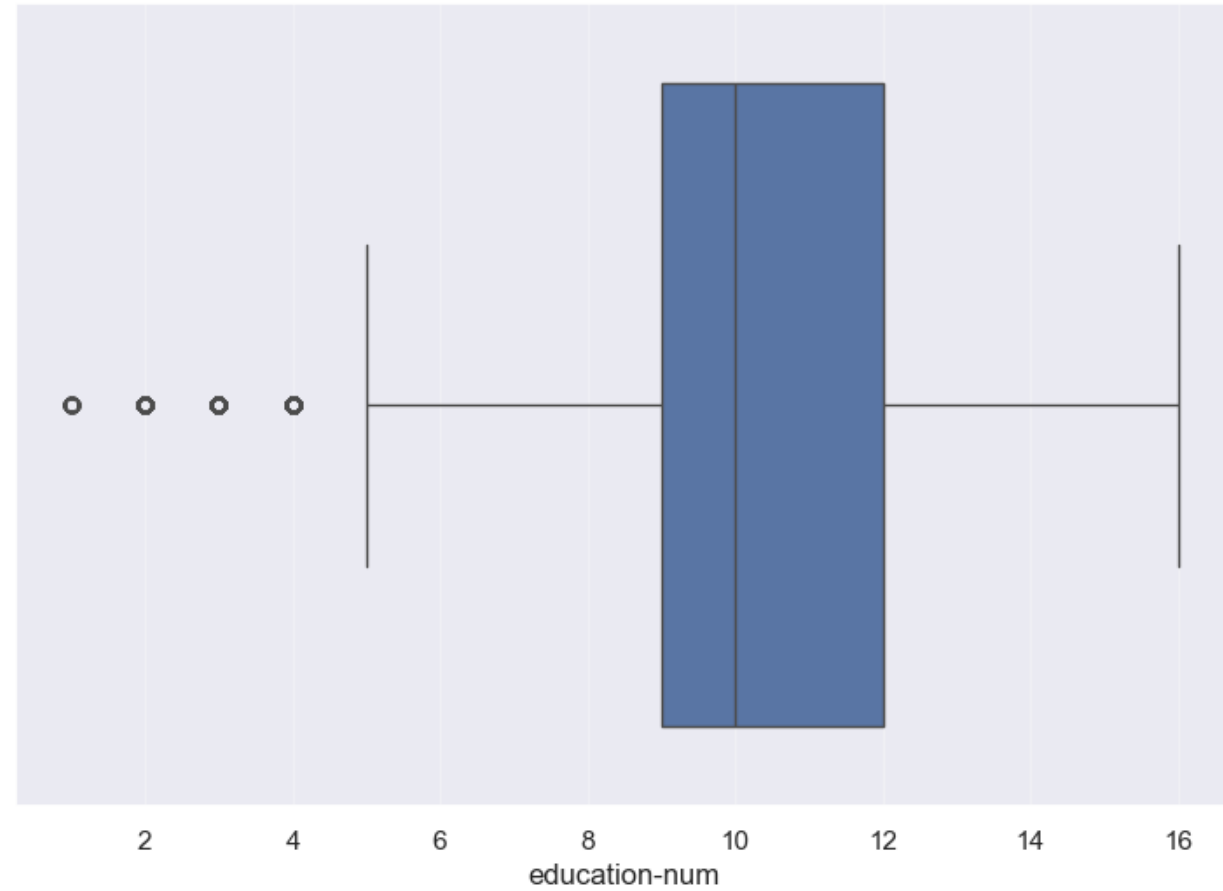
47.2% work exactly 40 h: box is only 40-45 h, IQR marks 26% as outliers

OUTLIERS

Box-and-Whisker Plot - age



Box-and-Whisker Plot - education-num

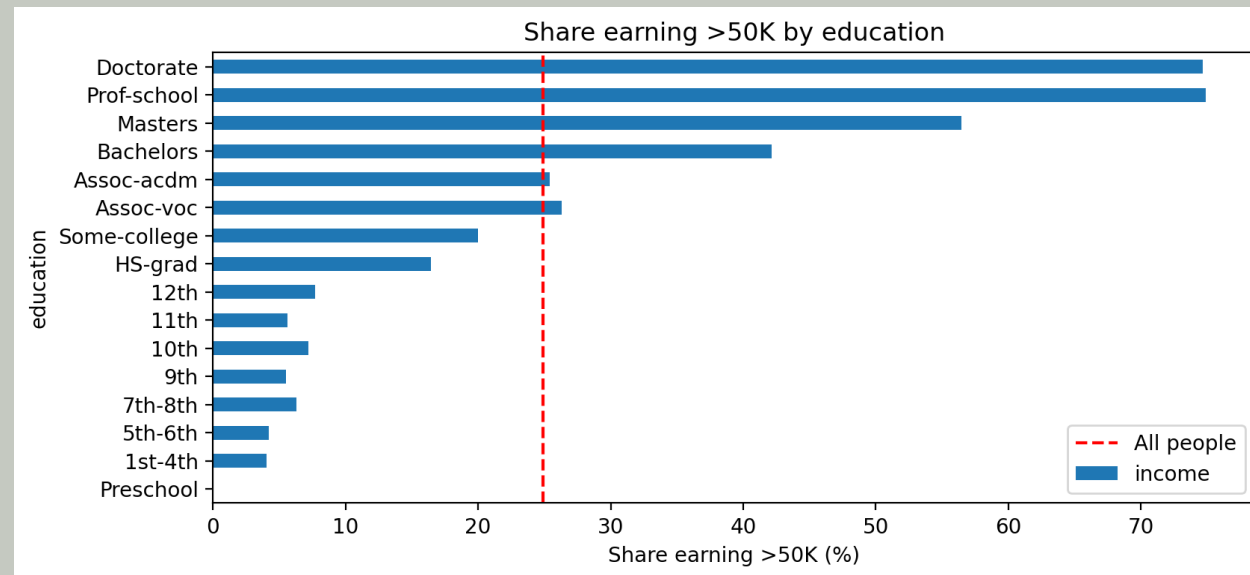


WHAT IS RELATED TO INCOME

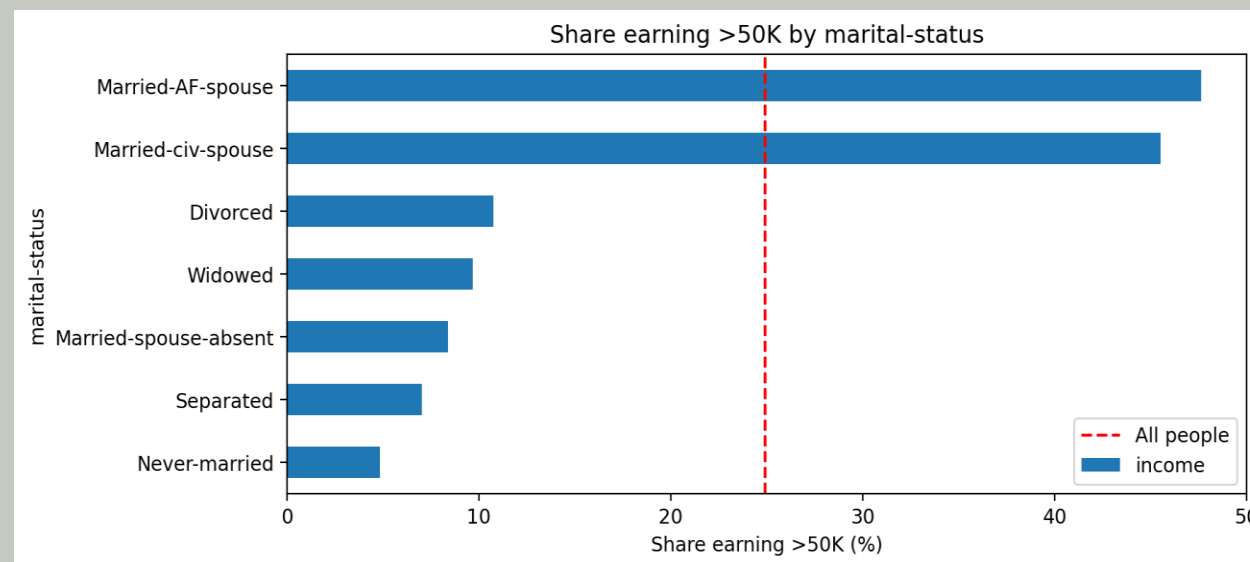
Married: 45.5% earn >50K

Others: 6.8%

Correlation with income (train)	r
Married-civ-spouse	0.45
education-num	0.34
has-capital	0.31
capital-gain-log	0.29
age	0.24
hours-per-week	0.23
Own-child	-0.23
Never-married	-0.32



No correlation above 0.45: no single variable predicts income, models must combine them



HYPOTHESIS TESTS

45 194 people (train + test), $\alpha = 0.05$

Test	H0	Result	Decision
One-sample t-test	mean working hours = 40	40.94 h, $p < 0.001$	reject H0
One proportion z-test	share earning >50K = 25%	24.80%, $p = 0.314$	do not reject
Pearson correlation	age and hours not correlated	$r = 0.10$, $p < 0.001$	reject (very weak)
Two-sample t-test	women and men work the same hours	36.9 h vs 42.9 h, $p < 0.001$	reject H0
Two proportions z-test	same share of women and men earn >50K	11.4% vs 31.3%, $p < 0.001$	reject H0
χ^2 goodness of fit	sex split 50/50	32.5% women, 67.5% men	reject H0
χ^2 independence	income does not depend on education	$\chi^2 = 5\,996$, $p < 0.001$	reject H0
χ^2 homogeneity	same income split in all race groups	$\chi^2 = 453$, $p < 0.001$	reject H0

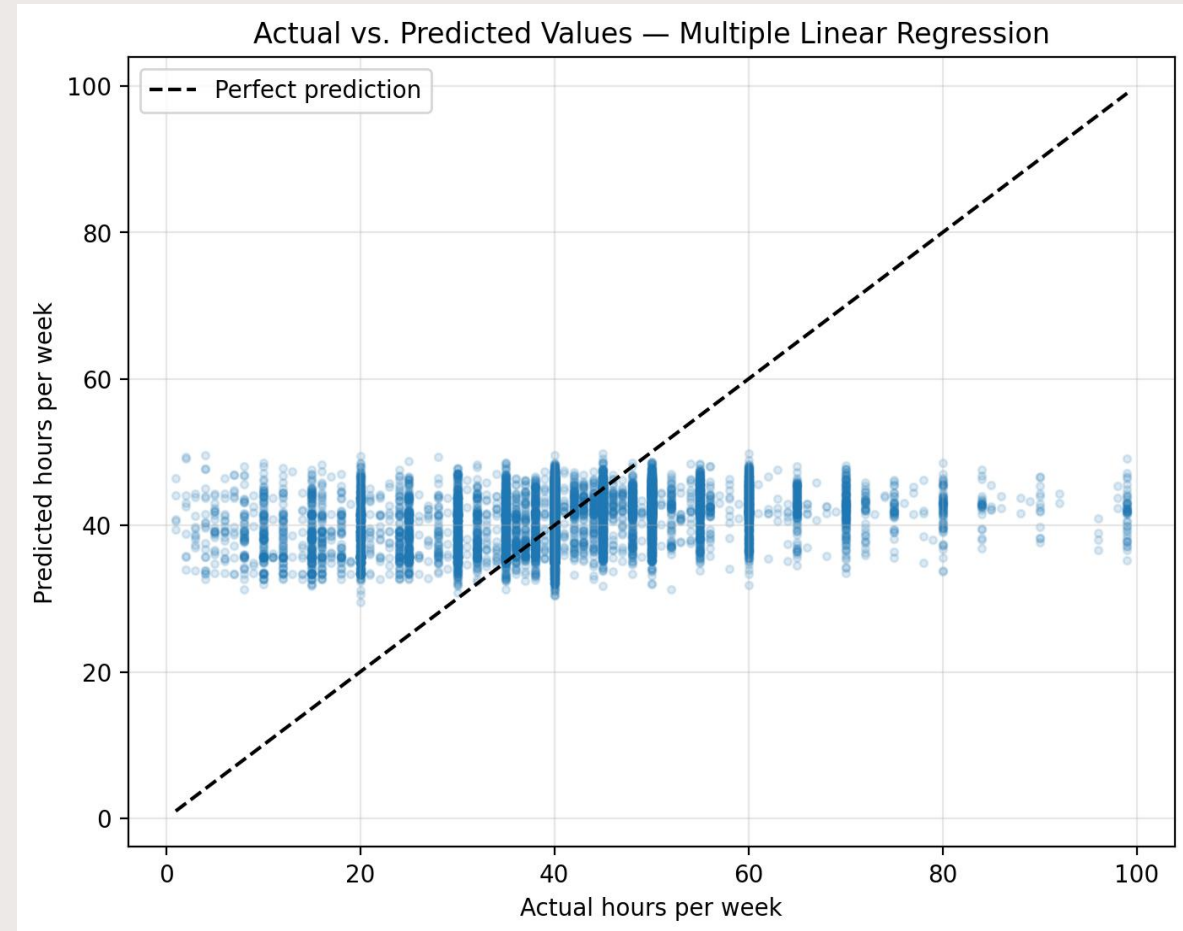
MULTIPLE LINEAR REGRESSION

Predicts hours-per-week; fitted on train, evaluated on test

$$\begin{aligned}\text{hours} = & 27.33 + 0.070 \times \text{age} \\ & + 0.696 \times \text{education-num} \\ & + 5.73 \times \text{male}\end{aligned}$$

Metric (test)	Value
MAE	7.74 h
MSE	134.14
RMSE	11.58 h
MAPE	30.9%
R^2	0.078

Significant (F = 899), but weak: explains only 8% of the differences

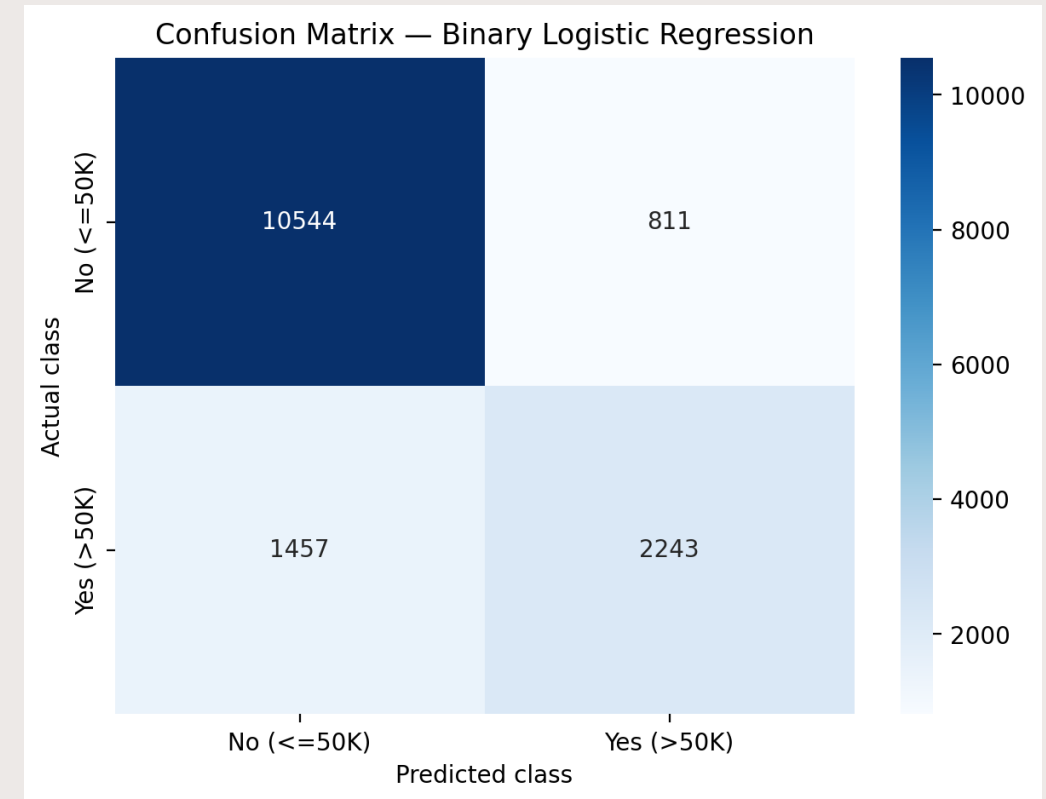


BINARY LOGISTIC REGRESSION

Variable	Odds ratio
Married-civ-spouse	×9.1
relationship: Wife	×3.7
male	×2.4
occupation: Exec-managerial	×2.2
each step of education-num	×1.32
relationship: Own-child	×0.54

Test set (baseline: always “≤50K”)

Accuracy	Baseline	Precision	Recall	F1	ROCAUC
0.849	0.754	0.734	0.606	0.664	0.905



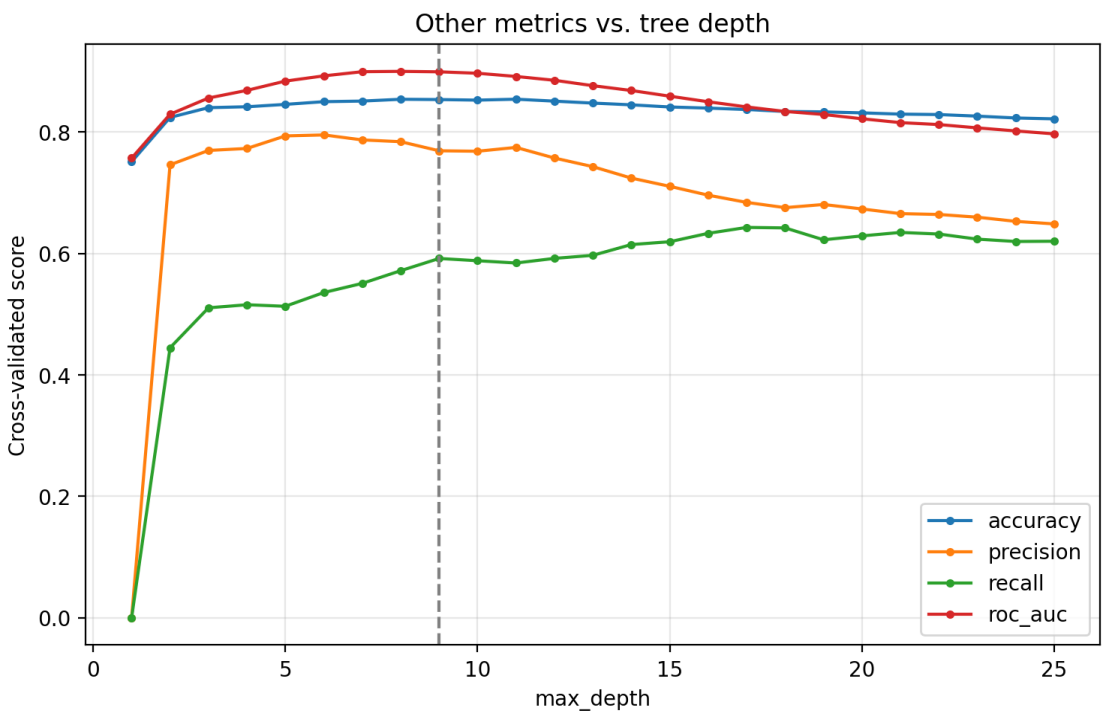
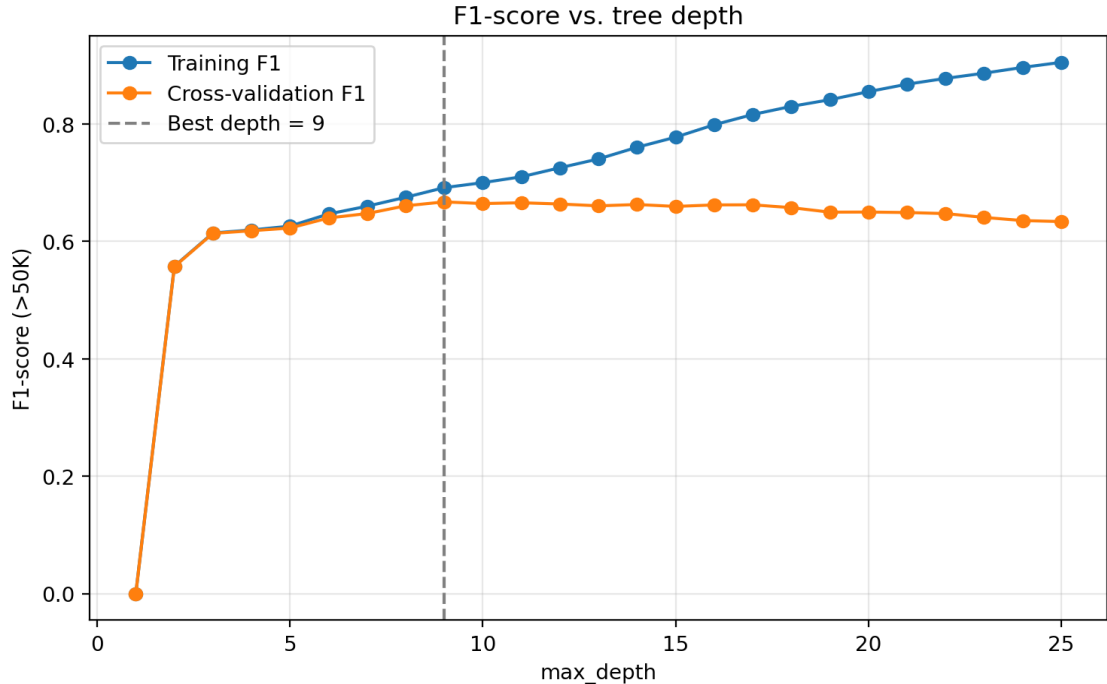
DECISION TREE CLASSIFICATION

Tree	Train accuracy	CV F1
No limits	97.7%	0.624
max_depth = 9	86.4%	0.667
min_samples_leaf = 20 (final)	87.2%	0.678

Settings chosen with 5-fold cross-validation on train. The test set is used only once and for the final tree.

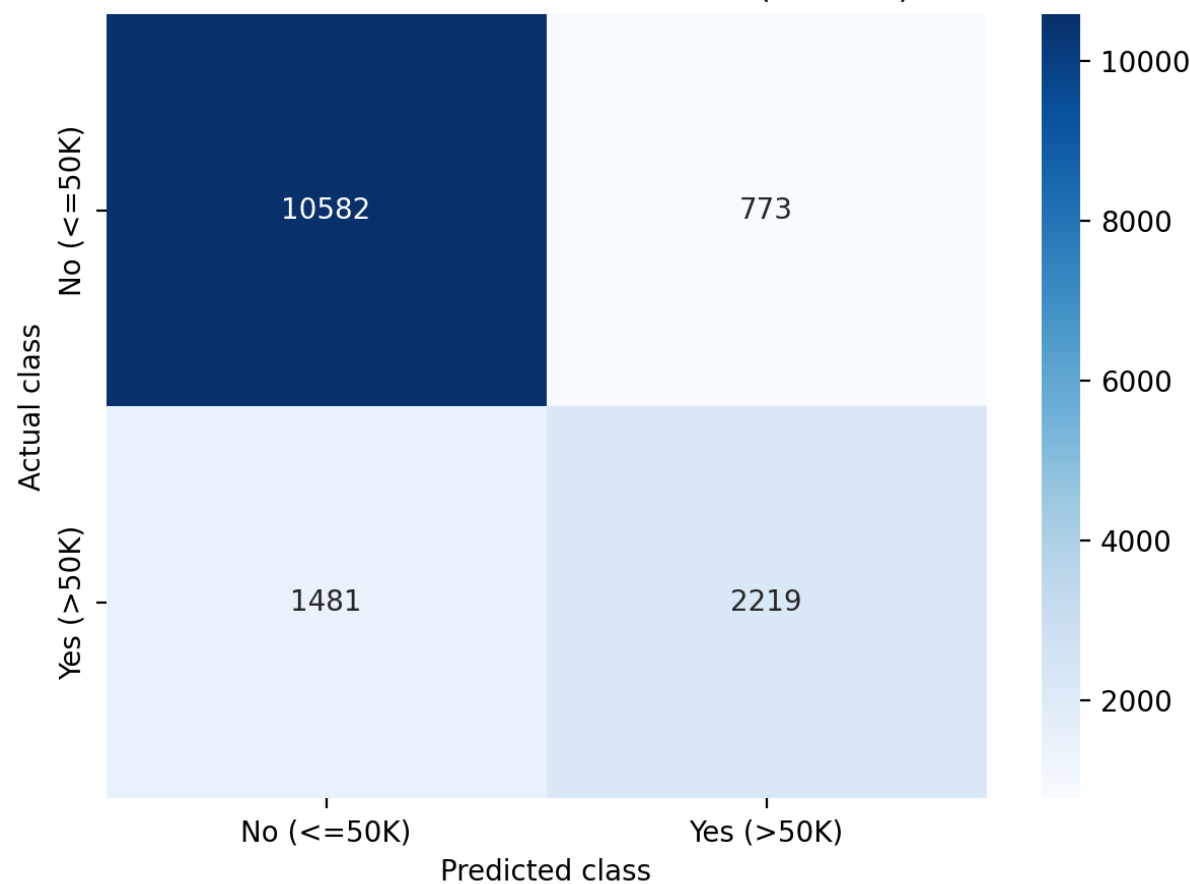
Final tree, test set

Accuracy	Precision	Recall	F1	ROC-AUC
0.850	0.742	0.600	0.663	0.896

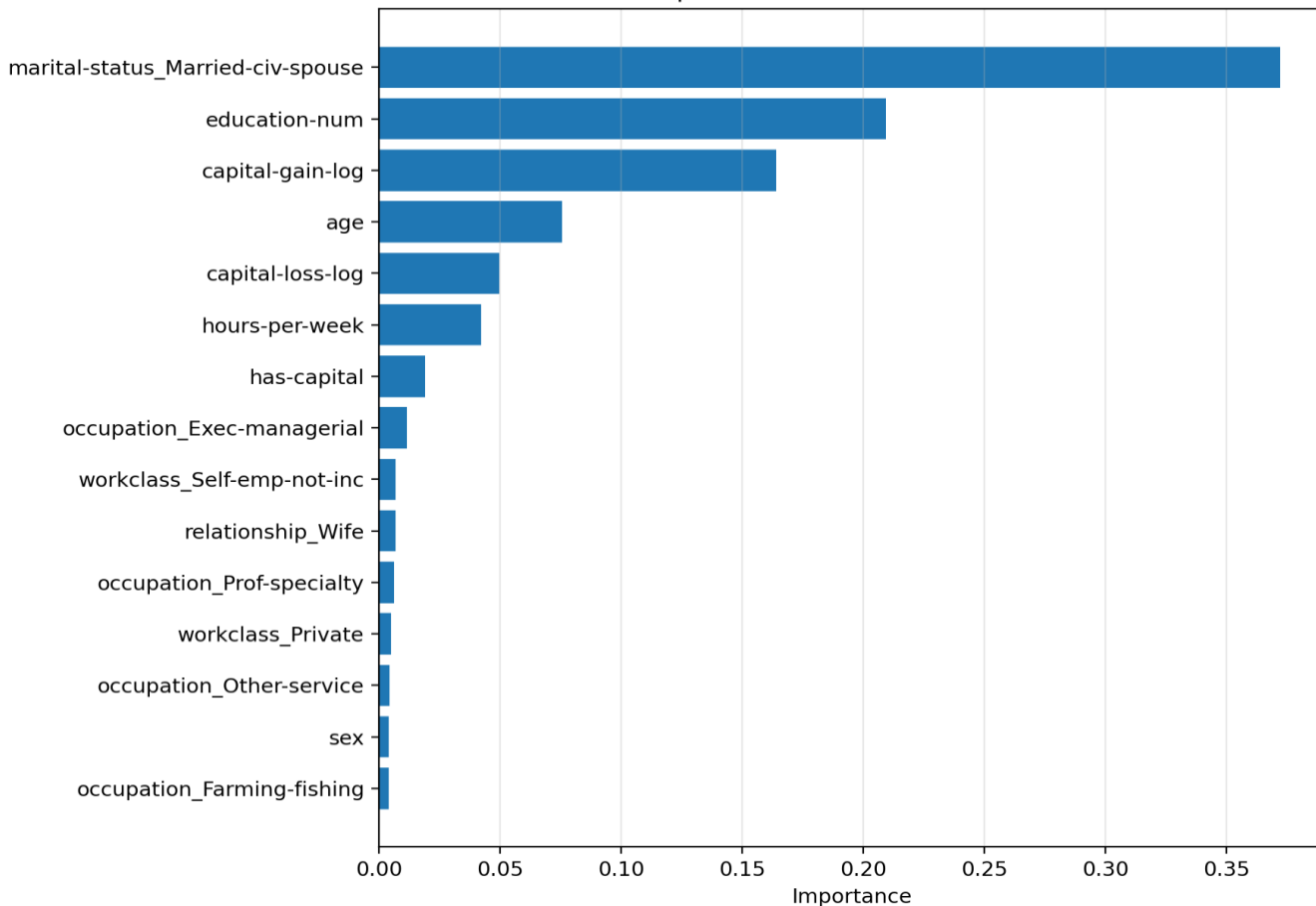


DECISION TREE CLASSIFICATION

Confusion Matrix — Decision Tree (test set)



Top 15 features — Decision Tree



MODEL COMPARISON

Test accuracy, same train/test split

Model	Accuracy
Always predicting $\leq 50K$ (baseline)	75.4%
Naive Bayes (published)	83.9%
C4.5 decision tree (published)	84.5%
Our logistic regression	84.9%
Our decision tree	85.0%
NBTree (best published)	85.9%

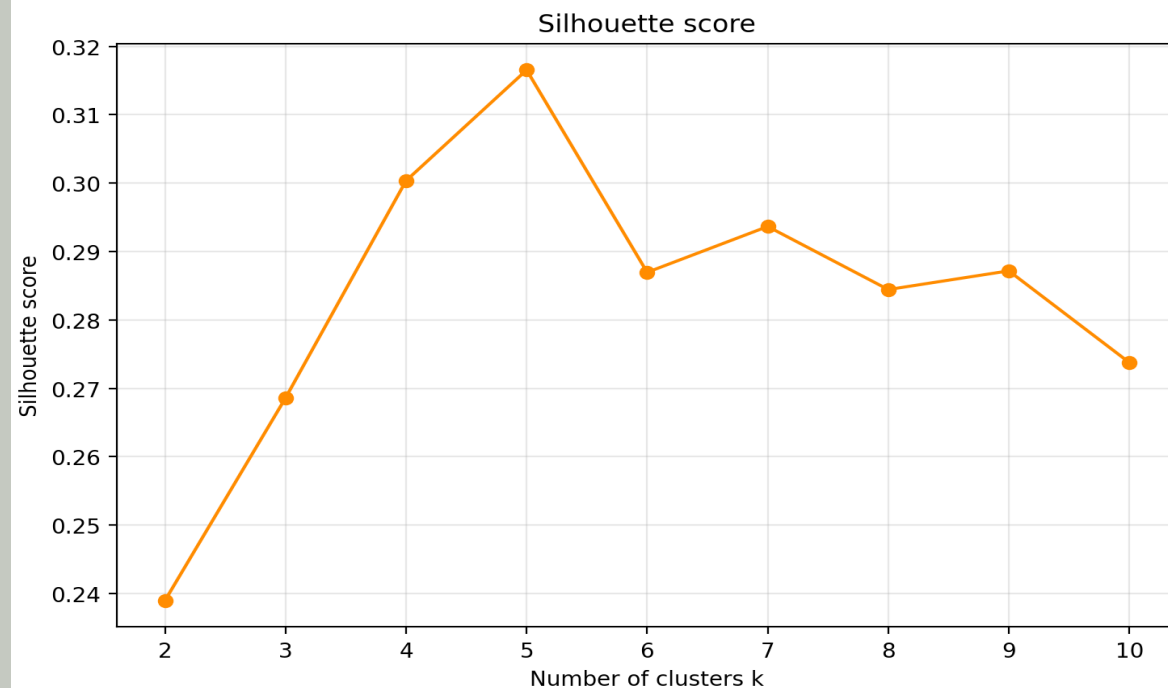
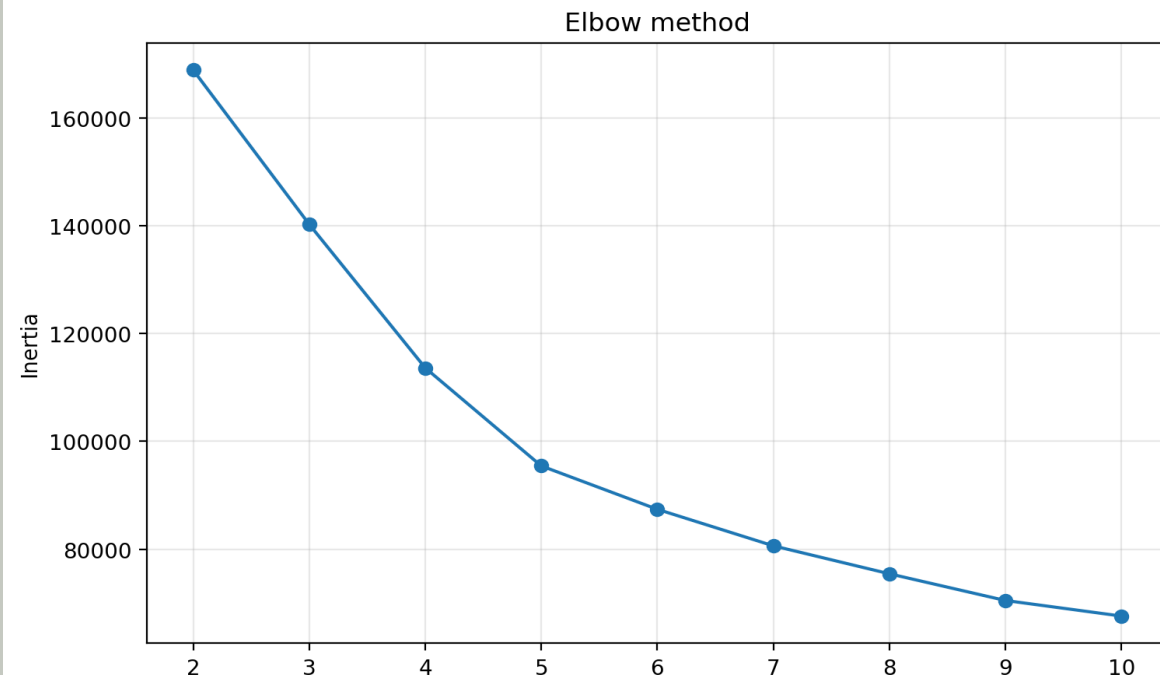
- Both models $\approx 85\%$, in the published range (84–86%)
- Two very different models, same result: the limit is in the data
- About 40% of high earners look like low earners in these variables

CLUSTERING (K-MEANS)

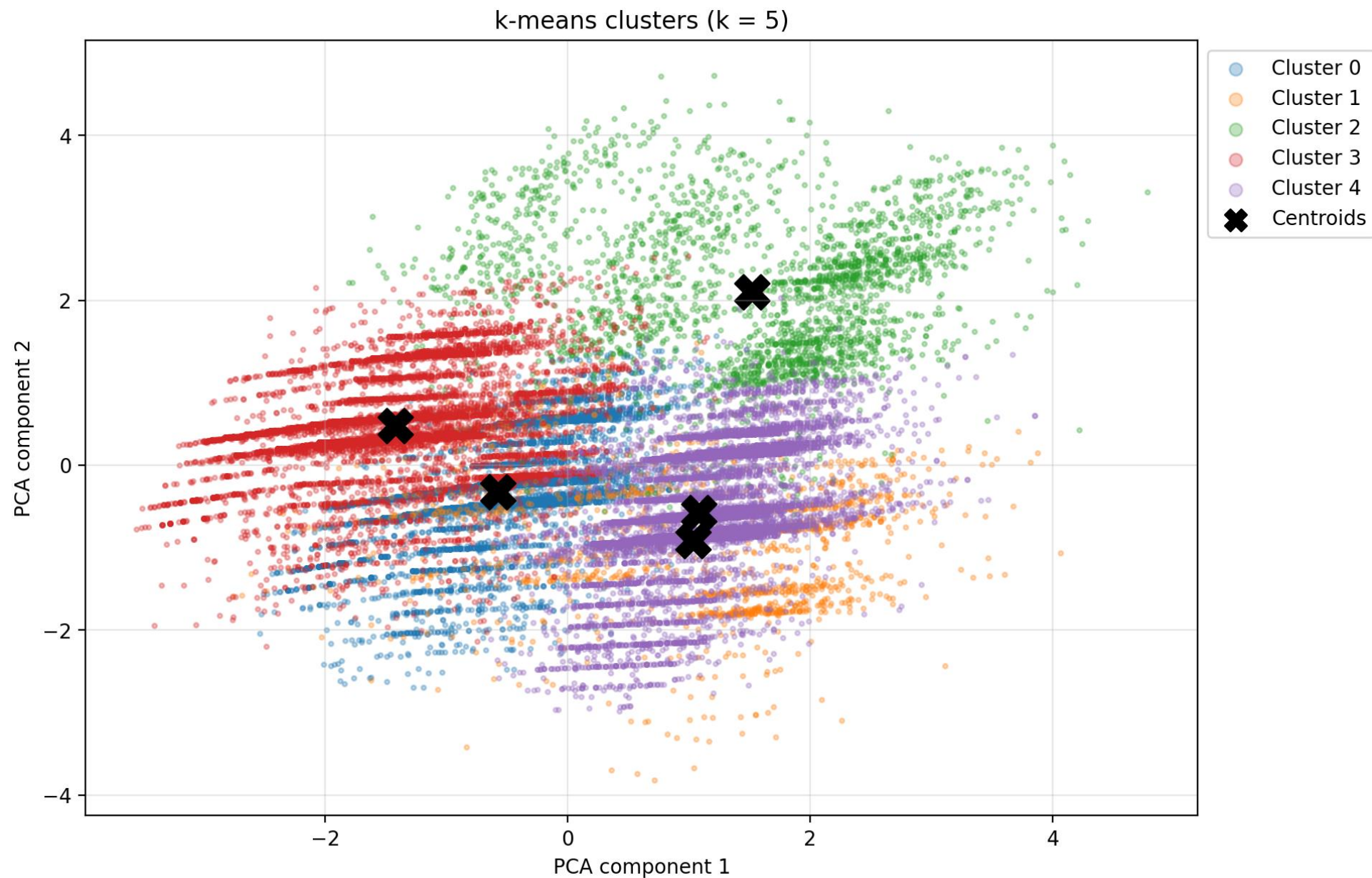
k-means on train, 7 standardized features (age, education-num, hours-per-week, capital-gain-log, capital-loss-log, sex, is-married). Income is not used.

Elbow method and silhouette (0.317) both choose $k = 5$.

Cluster	Who	Share	Earn >50K
0	Young single men (~ 32)	23%	6.1%
1	Capital loss (~ 42)	5%	51.6%
2	Capital gain (~ 44)	8%	63.8%
3	Women, mostly not married	29%	8.2%
4	Married men (~ 43)	35%	38.6%



CLUSTERING (K-MEANS)



CONCLUSIONS

Being married, education, capital gains and age are the factors most related to income in every part of the analysis.

Men work about 6 hours more per week and earn more than \$50K almost three times as often as women (31.3% vs 11.4%).

Our logistic regression (84.9%) and decision tree (85.0%) predict income almost as well as the best published model (85.9%), while working hours cannot be predicted well ($R^2 = 0.078$).

k-means, without using income, found 5 groups whose share of high earners ranges from 6% to 64%.

Original train/test split kept, settings chosen on train only: the test results are honest.

About 40% of high earners are still missed: a limit of the census data.